MALWARE – PART I

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Content

- Definition & History
- Malware Classification
- Malware Communication
  - Communication architecture
  - Communication Channels
  - Resilience / Stealth

Goals

- You know the most important «types» of malware like worms, Trojans, ransomware, rootkits, bootkits,…
- You know several techniques how malware communicates and how malware hides itself from the prying eyes of users and defenders
- You have a basic understanding of why our defences against malware are still quite weak

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Malware - Overview

Definition

- Malware is an acronym for **mal**icious soft**ware**
- It comes in different formats; executables, shell code, scripts, firmware, …
- Two definitions:
  - NIST [1]: A program that is inserted into a system, usually covertly, with the intent of compromising the confidentiality, integrity, or availability of the victim's data, applications, or operating system or otherwise annoying or disrupting the victim.
  - OR-MEIR et al. [2]: Malware is code running on a computerized system whose presence or behavior the system administrators are unaware of; were the system administrators aware of the code and its behavior, they would not permit it to run. Malware compromises the CIA of the system by exploiting existing vulnerabilities in a system or by creating new ones.
- Most definitions include at least one of the following two main traits:
  - maliciousness or harmfulness
  - the ability to perform actions without the user's consent or knowledge

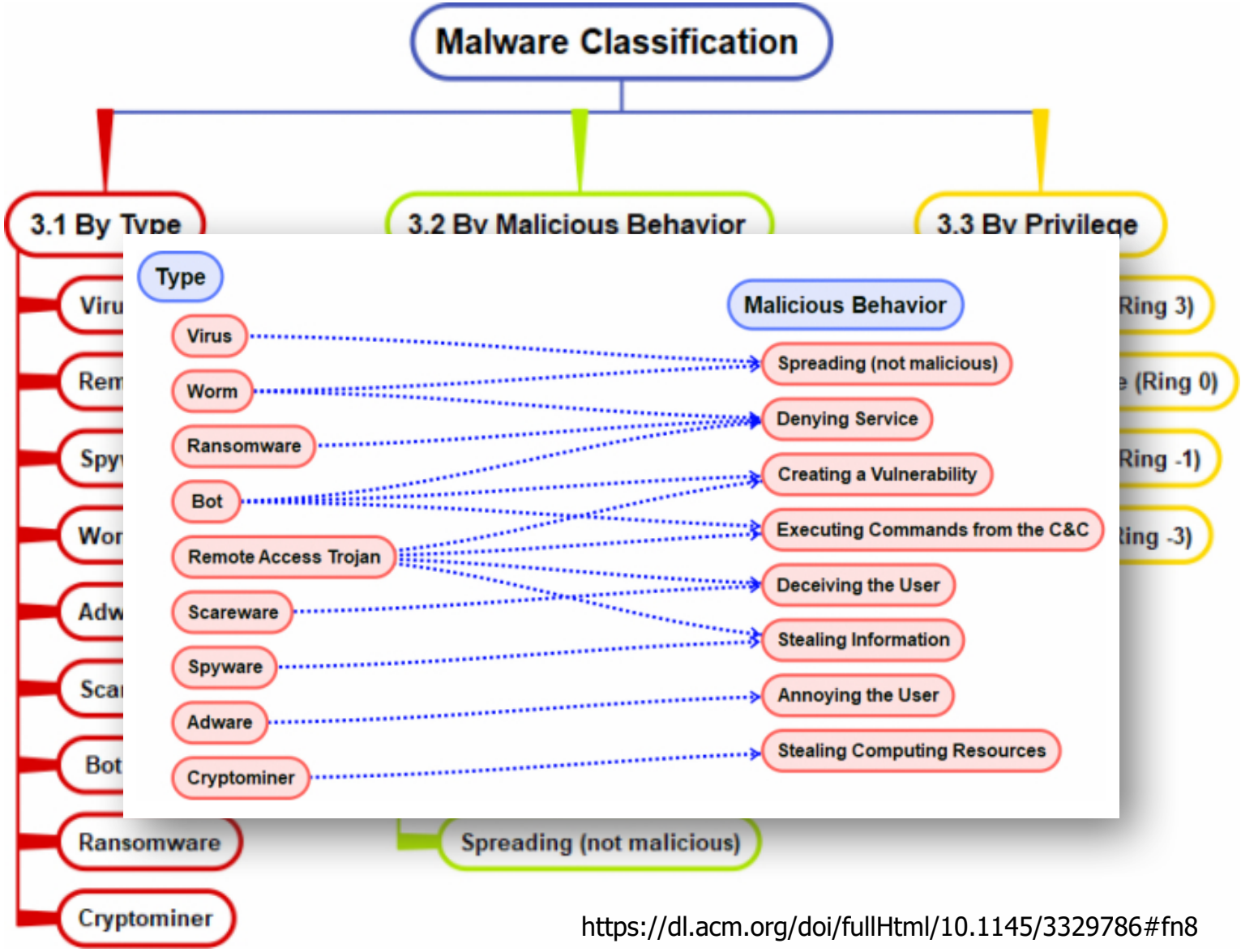
A Brief History of Malware

- 1949 – John von Neumann describes a design for a self-reproducing computer program => theoretical father of computer virology
- 1982 – Elk Cloner, the first computer virus in the wild
  - Written for Apple II by Rich Skrenta, a 15-year-old high school student
  - Boot sector virus spreading to other disks inserted when resident
  - Displayed a poem on every fiftieth boot from an infected disk
- 1988 – Moris Worm, the first computer worm distributed via the Internet and to gain mainstream media attention
  - Written for Unix by Robert Morris, graduate student at Cornell University
  - Infected 2'000 computers in <15 hours by exploiting vulnerabilities in sendmail, finger, and rsh/rexec, as well as weak passwords
  - Should do no harm but multiple infections of the same machine slowed them down to a point of being unusable
- 2001 – Win32.5-0-1, the first social networking virus (MSN Messenger and bulletin boards) written by Matt Larose
  - Required a click to get infected and sent out an email with user data

Malware Classification

- Malware classification helps us in multiple ways, for example to have a common language and understanding of a malware and for risk scoring
- There are multiple types of classification with different end goals
  - By type: Based on the main features implemented by the malware (e.g., replication method)
    - Terms frequently used in mass media, for example bot, virus or worm
  - By behavior: Based on the exhibited behavior
    - For example, "information stealer"
  - By family/lineage: Focus on the authorship and the lineage / evolution
    - For example, by analyzing the code-case for shared or similar code
- The problem with malware classification: there is no generic commonly agreed classification scheme
- For the types, there is some consensus on typical characteristics of a malware of that type, but details might vary

Malware Classification



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Malware Types & Terms

- There is no such thing like mutually exclusive malware types but rather categories of malware that are not mutually exclusive
- Aside from the types below, you should also know what these terms mean: Living Off the Land and Fileless Malware

- Trojan Horse
- Backdoor
- Remote Access Tool
- Downloader
- Dropper
- Bot/Botnet
- Monitor / Spyware
- Mailer Scareware
- Adware
- Ransomware
- Cryptominer
- Spyware
- Rootkit / Bootkit
- Worm
- Virus

Trojan Horse

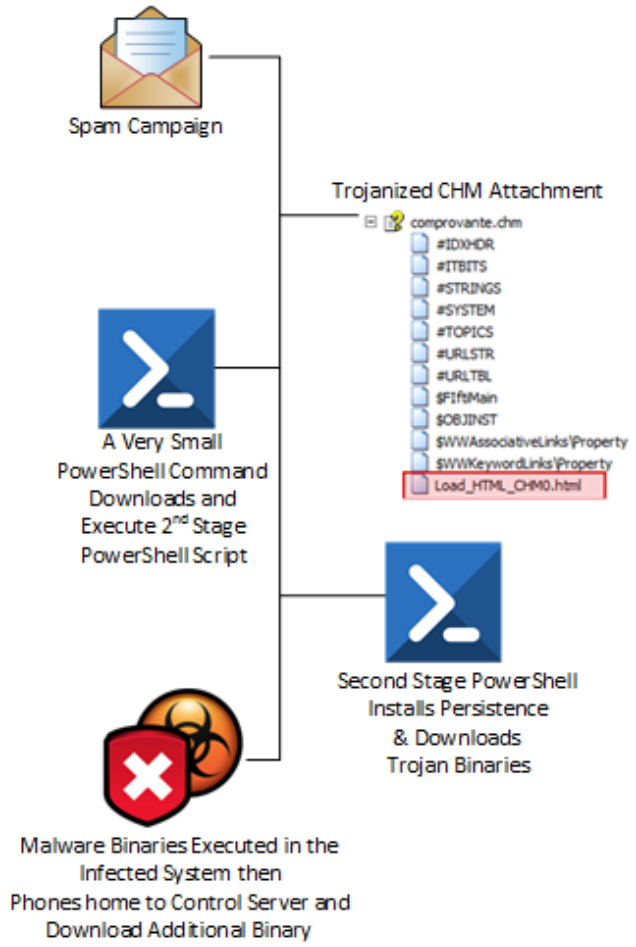
- Trojan or Trojan Horse is frequently used to classify malware samples that employ some mechanism to conceal their true identity.
- Some examples:
  - A whitepaper you downloaded concealing a PDF exploit
  - A binary version of a 3rd party library containing a backdoor
  - Malware embedded within a bundle that pretends to be a printer driver
  - A music player that contains embedded backdoor code when you run it

Backdoor & Remote Access Tools (RAT)

- Traditionally, backdoor functionality provides some level of interactive 1-to-1 access to a compromised system
  - Usually provided through a network connection between adversary and the target's host.
- A Remote Access Tool is frequently used to describe a backdoor having a high level of interactive functionality and providing expansive access to the target's host.
  - Sometimes, the level of access provided is more powerful than the normal end-user interface of the targeted system.

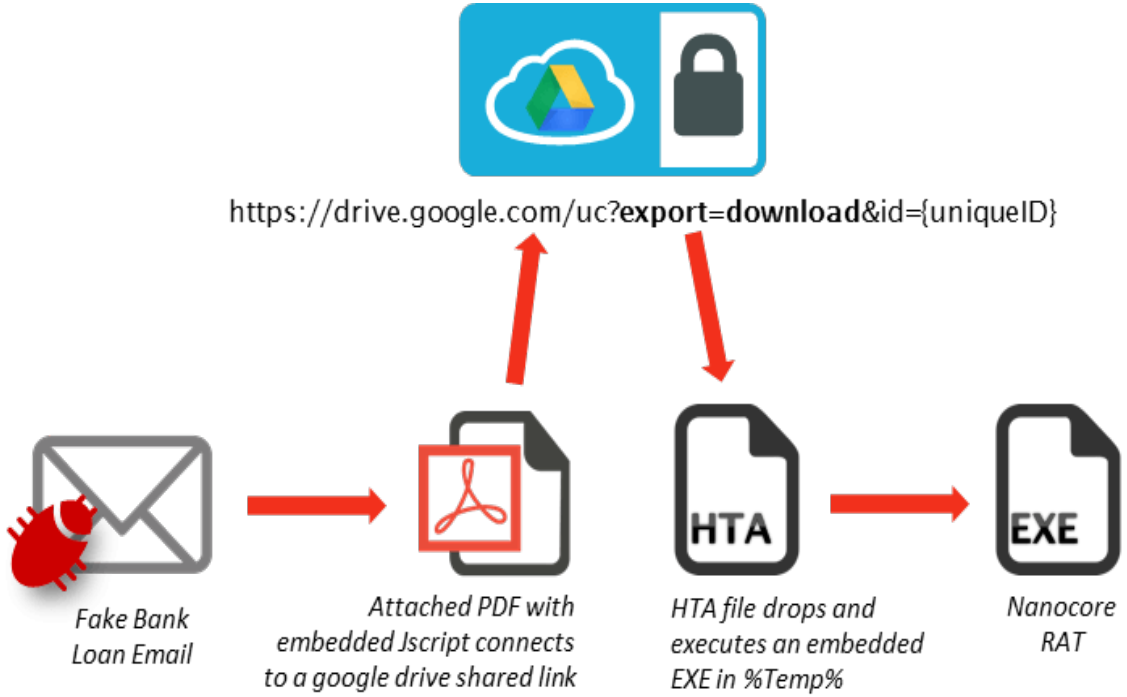
Downloader

- During a mission, attackers frequently employ multiple tools
- Downloader functionality provides a tool with a specific feature to download other tools
- In the initial phase (e.g., spear- phishing), adversaries often deliver a lightweight tool
- It may be pre-configured to pull down more feature-full malware
  - Attributes of the compromised host may guide from where and what to retrieve next
- The breaking-up of attacks helps to reduce detectability and facilitates surgical retooling.



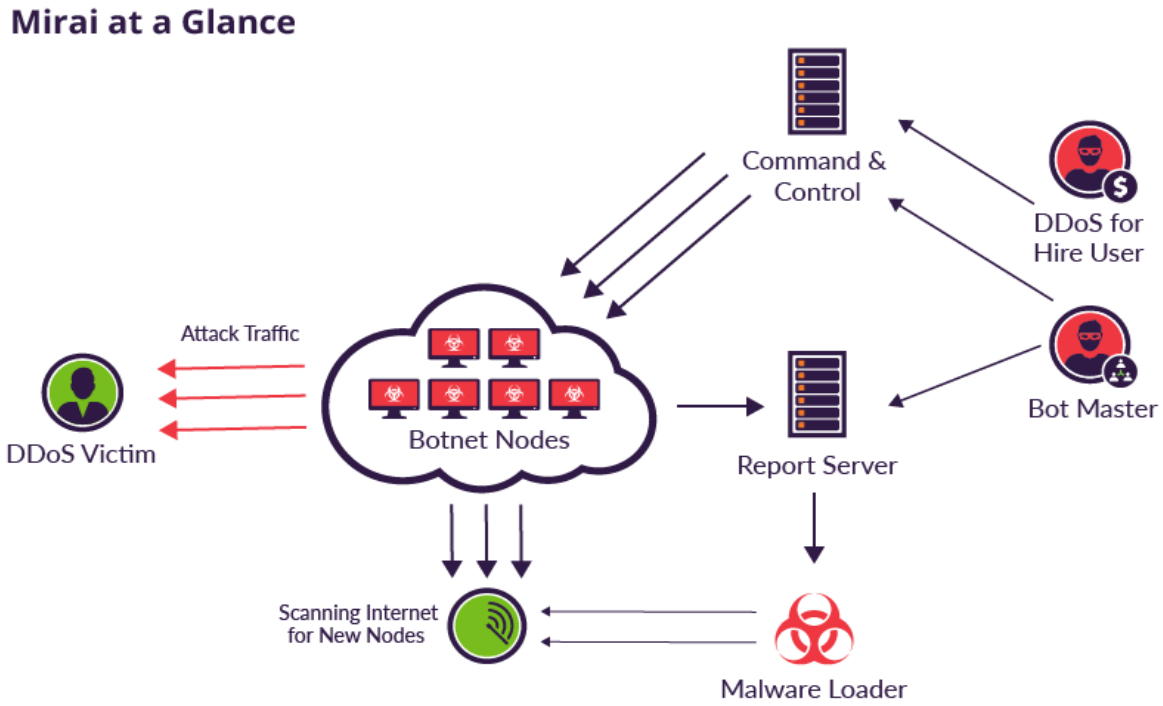
Dropper

- In a multi-stage attack, one tool may need to write/drop another one to disk and execute it.
- Code must first be extracted and stored in an OS-compatible, native container (e.g., EXE file).
- Example:
  - HTA document that contains embedded, encoded malware
  - When opened, the embedded malware is extracted, dropped to disk and executed
- **Note:** Downloaders are usually also Droppers, but not vice versa



Bot / Botnets

- Botnet - A number of interconnected computers (=nodes) running a bot
- The bot is coordinating their actions automatically or by command and control (C2/C&C) from the "operator"
- Similar to Backdoor but control is usually 1:m instead of 1:1
- Common use cases for a botnet and the power of coordinated actions are:
  - Distributed Denial of Service (DDoS)
  - Spam campaigns
  - Stealing credentials at scale

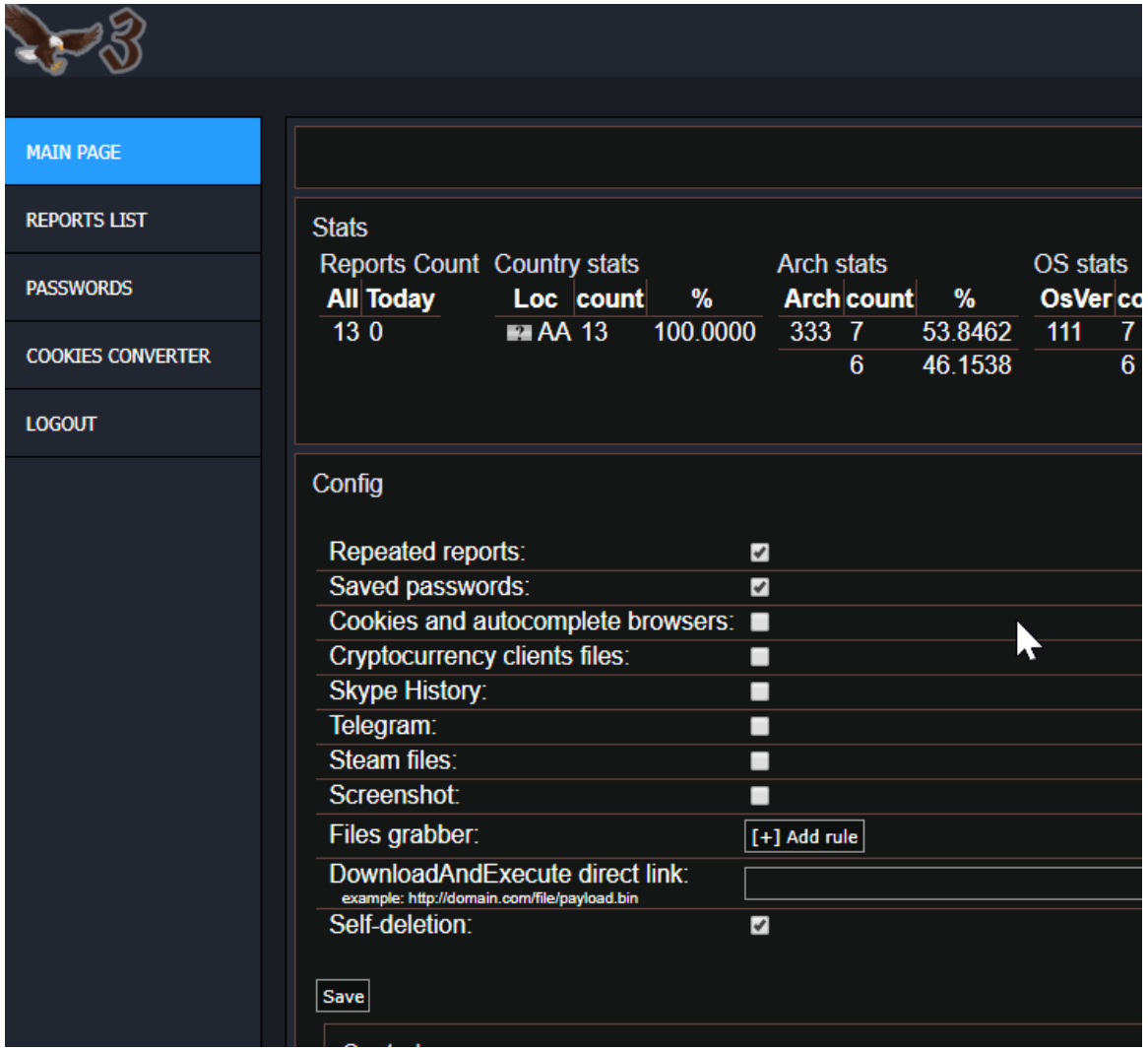


Spyware (Monitor)

- Malware with mechanism to record user activity or the environment employs monitor functionality
- The recorded data is delivered back to the adversary
- Example monitor sources on a target:
  - Webcam / Microphone
  - Desktop recording
  - Password prompts
  - Common data folders (My Documents, …)
  - Keyboard (keylogging)
  - Web browser traffic
  - Network traffic

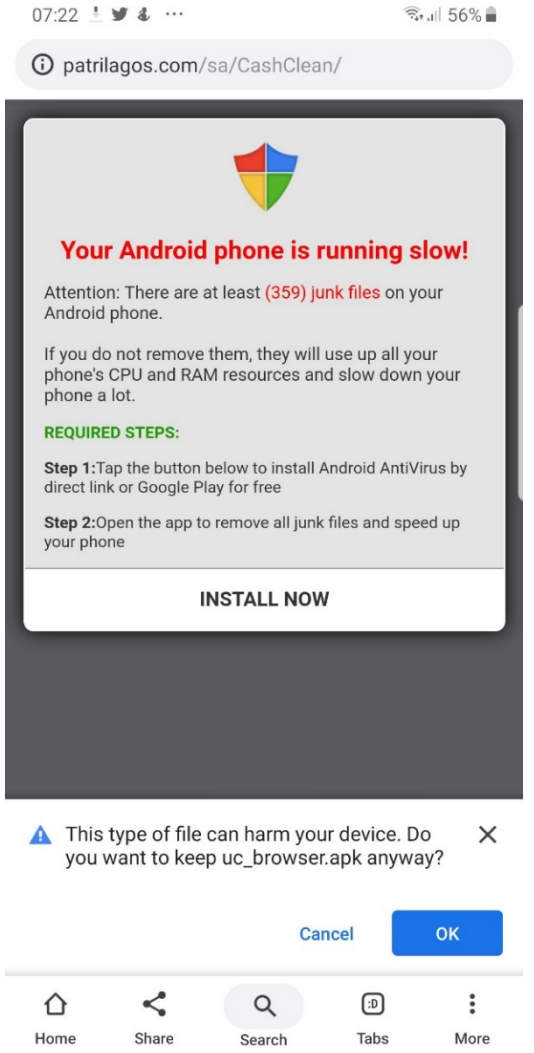
Information Stealer (Spyware)

- Malware with functions to automate the process of stealing information
- It searches data using a hard-coded or network-provided collection plan
- Common approaches include:
  - Contact list theft
  - Browser cookies/history
  - Targeted document theft
  - Credential- / Keystores
- Similar to Monitor but with automated search & extraction of relevant data only



Scareware / Adware

- Employed to achieve a social engineering outcome
- Entices or frequently scares the user into taking specific actions
- A common variant is “Fake Anti Virus” programs
  - When visiting a website, visitors are told they have a virus
  - The site offers a (fake) free Anti-Virus program to remove it
  - It then annoys the user (e.g., with popups) until a fee is paid
- Adware is another common variant where the software may not be an illegal enterprise, but is annoying nonetheless



Ransomware

- Similar end-goals as scareware - social-engineer the target into paying a fee or some other activity
- Typically, ransomware encrypts files on the local system and asks for money to decrypt them
  - Contemporary ransomware leverages strong public-key cryptography
  - The malware only contains the public key => makes it practically impossible to recover the files without paying

Virus and Worm

	Virus	Worm
What does it do?	Inserts malicious code into a program or files that can contain «executable code» (e.g., macros, scripts,…)	Exploits a vulnerability in an application, service or operating system*
How does it spread to other computers?	Users transfer infected files to other computers. They are then executed/ used there	Uses a network to travel from one computer to another
Does it infect a file?	Yes	No
Does it spread with help of humans?	Yes	No
Example	Sality	WannaCry

- Virus and worm functionality describes approaches to self-propagating/replicating malware (example: WannaCry)
- In practice, distinction might be hard (e.g., email-worm/virus)

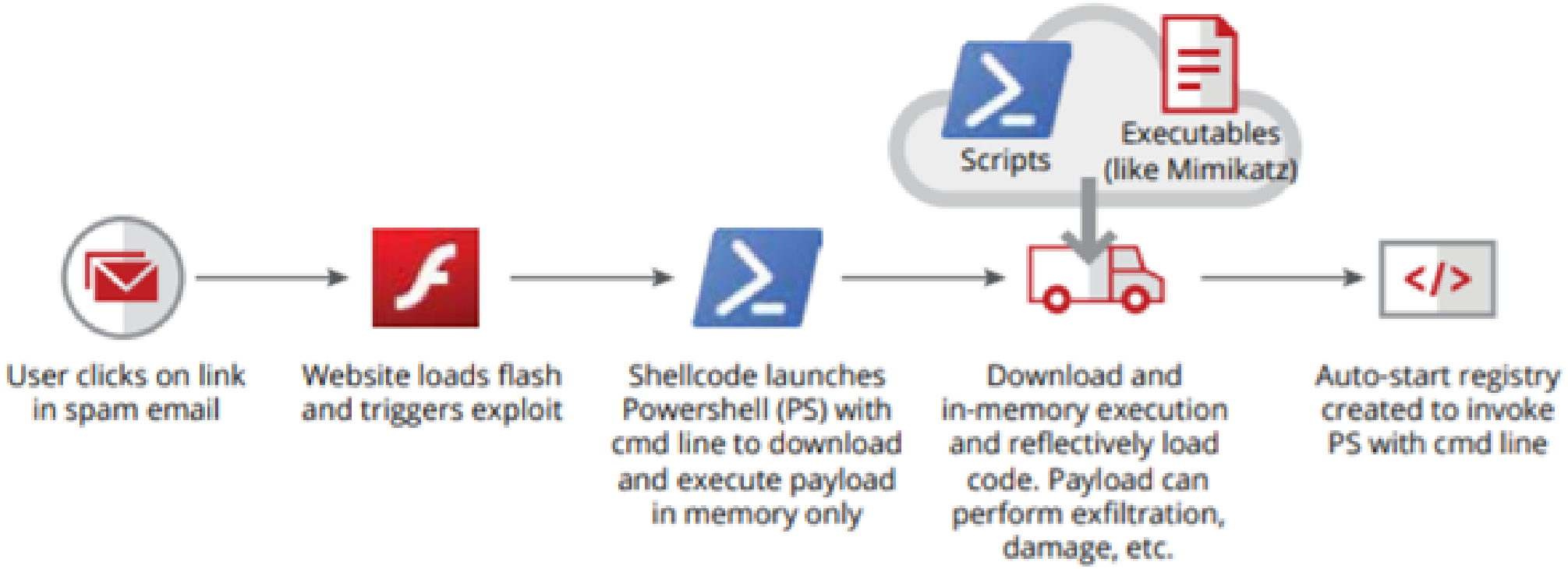
Famous Computer Worms

- Email Worms – Malicious attachment or link to infected website
  - ILOVEYOU worm hit 10% of Internet connected computers (50 million infections in 12 days) and caused >15 billions of estimated damage
- Internet worm will scan all available network resources using local operating system services and/or scan the Internet for vulnerable machines to connect and gain access to them.
  - Famous examples: Code Red, Blaster, Witty, SQLSlammer, Conficker
  - Witty worm: One UDP packet to rule a security product (ISS firewalls)
- Fast-spreading worms (without requiring user interaction)
  - Scanning for infected hosts (often, local network first, then the Internet)

Other Terms and Types in brief:

- Mailer (or Spambot) – Malware with functionality to send emails
  - Deliver emails directly via SMTP or using web-based mail accounts of the victim
- Cryptominer / Cryptojacking – Malware with functionality to mine crypto currencies. Impact is usage of your resources for free.
- Living Off the Land - Malware that utilizes already installed software to implement some/all of its functionality (e.g., PowerShell or a legitimate remote management software)
- Fileless Malware – No stand-alone malware is installed
  - Malware code is only in memory, not on disk/in files
  - In-memory only: Exploit vulnerability in a software to inject the malware into that process from remote without any file-system access
  - Software that is already installed is leveraged to load malware code into memory (subset of living off the land)

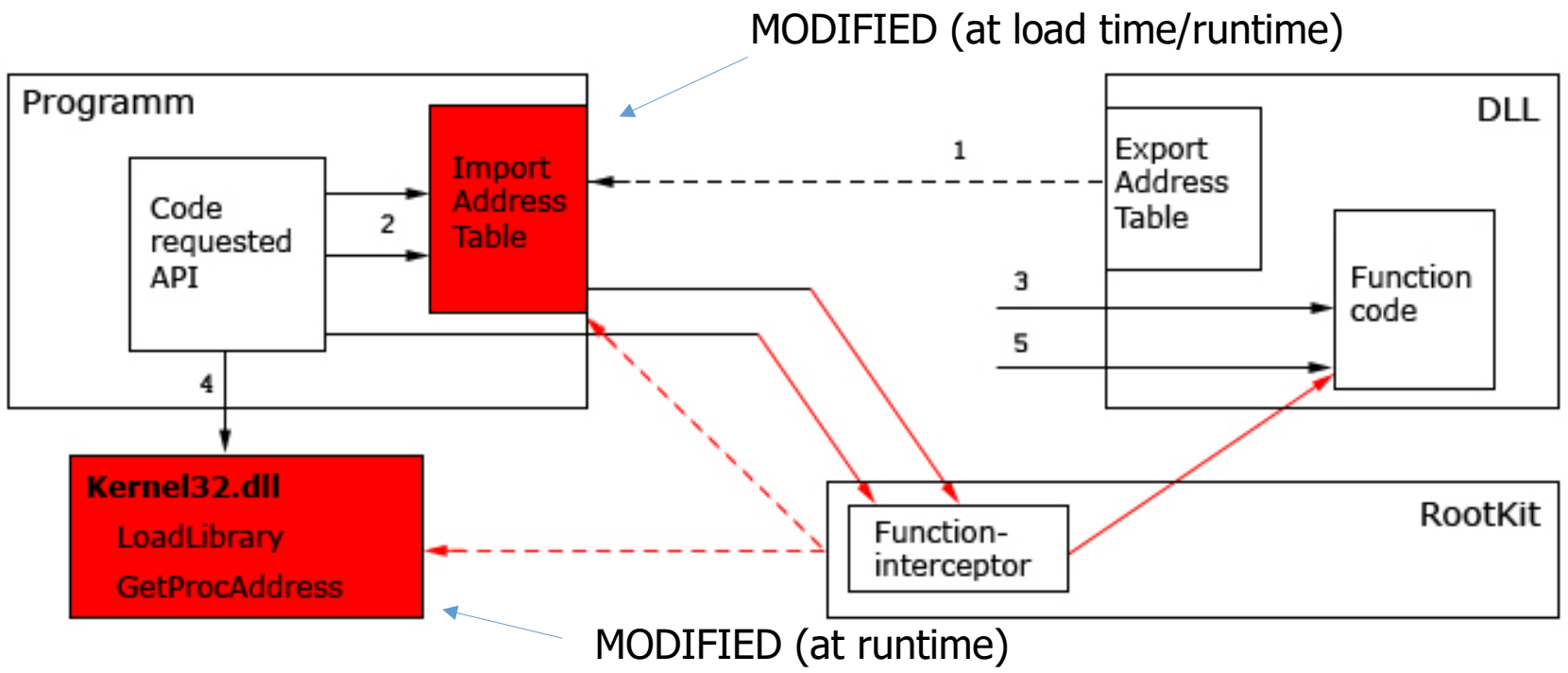
Fileless or not Fileless?



Rootkits

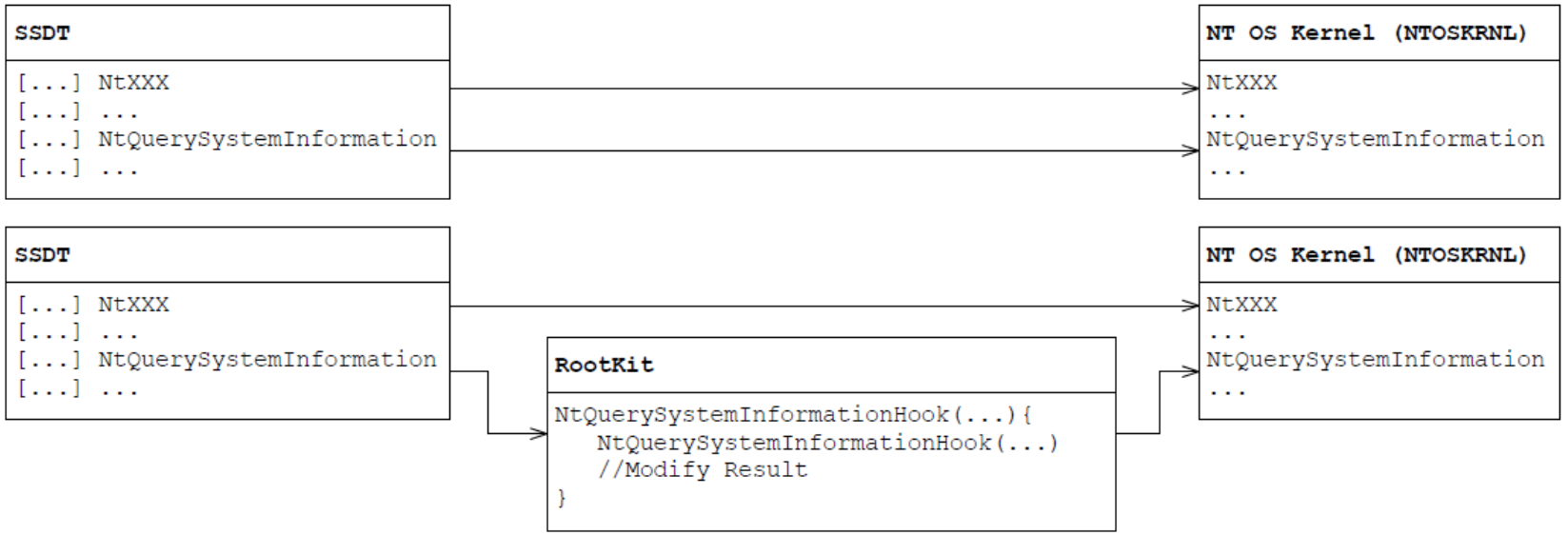
- Traditionally, rootkit functionality utilizes operating system interfaces to conceal the presence of malware (=stealth)
- Traditional rootkit variants (next slides): User-mode and kernel-mode rootkits
- Other rootkit variants:
  - Bootkit - runs in real mode (and kernel mode after booting)
    - Infects startup code (boot sector) to take control of the boot process
    - Can subvert the kernel and its security protections
  - Hypervisor level rootkits
    - Exploits hardware virtualization features such as Intel VT or AMD-V
    - Runs in Ring -1 and hosts the target operating system as a VM
    - Does not have to make any modifications to the kernel of the target
  - Firmware and hardware rootkits
    - Run on routers, network card, hard drive or system BIOS

Rootkits – User Mode



- Run in user space and work best with administrative privileges
- Alter security settings and hide processes, files, system drivers, network ports, and even system services
- Hiding techniques: E.g., function hooking of widely used API in every process

Rootkits – Kernel-Mode



- Loaded as device drivers / loadable kernel modules => requires to be able to install a driver (kernel module)
- Many different hiding techniques
- E.g., SSDT (System Service Descriptor Table) hooking

Rootkits – Kernel-Mode

- Protections:
  - Windows Driver Signing – Starting with Windows 10, drivers must be submitted to, checked and signed by Microsoft
  - PatchGuard (all x64 Win OS since XP) protects from SSDT-like rootkits by (trying to) preventing modifications of data structures like the SSDT
  - Device Guard locks a device down so that it can only run trusted applications

- Device Guard:
  - Available on Windows 10 Enterprise or higher
  - Operating system only runs code that's signed by trusted signers
  - Who's a trusted signer is defined by your Code Integrity policy through specific hardware and security configurations

Detecting Rootkits

- Based on the «concept» used for hiding
  - E.g., look for alteration of the SSDT
- Detection of sophisticated rootkits at ring 0 or lower is difficult
- Timing analysis as a generic detection method for rootkits
  - Hooked/modified operations show a different timing behavior
  - Baseline to compare to is the challenge
  - In case of hypervisor level rootkits, external time sources must be used

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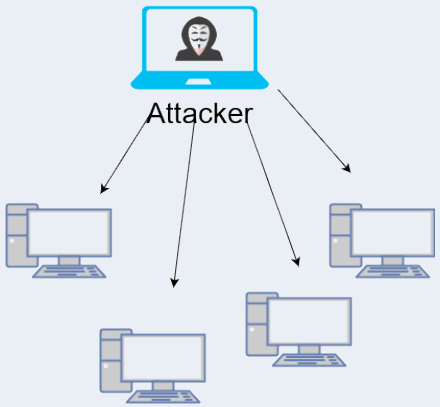
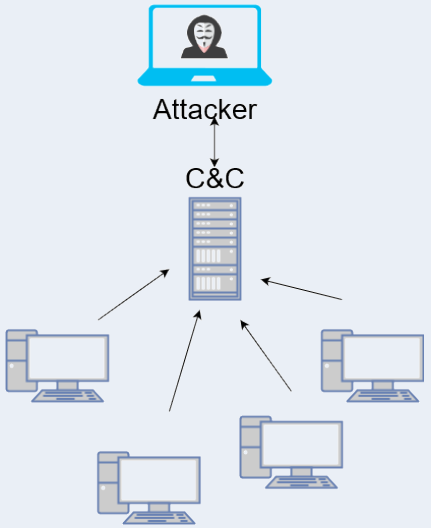
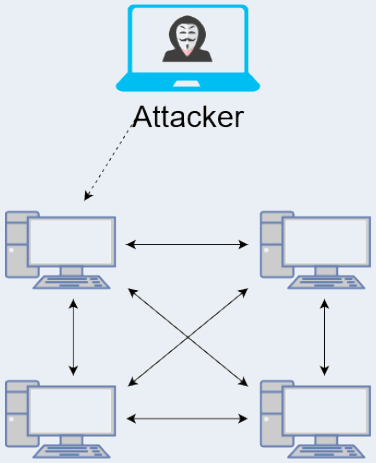
Malware Communication

- Many malware types need to communicate with the attacker(s) or other malware instances to fulfil their purpose
- Communication consists of the communication architecture and the properties of the communication channel itself
- Important goals for malware communication:
  - Communication should be resilient vs. blocking/takedowns
  - Communication should be hard to detect and/or identify

Communication Architectures

	Direct	Client-Server	P2P
Communication initiated by	attacker	malware	attacker/malware
Firewall compatibility	low	high	low-medium
Resilience / Ease of takeout	easy [hard]	easy [hard]	hard
Architecture Outline			

Client-Server – Mitigations of Single Point of Failure

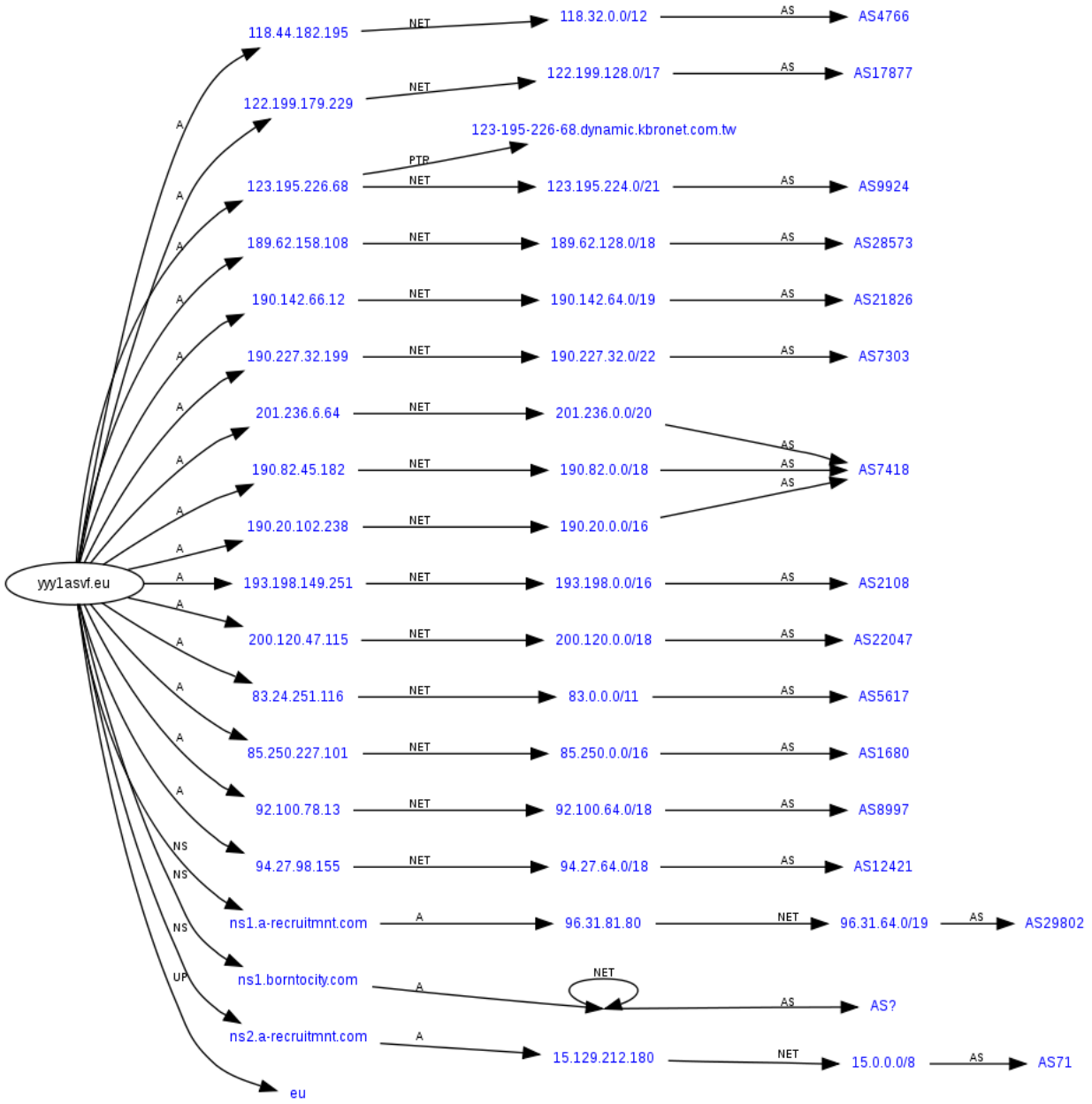
- If the IP/domain/URL of the C2 server is known, we can take it down (e.g., arrest the attacker), block connections to it or sinkhole* it
- Some mitigations:
  - Change the IP(s) on a regular basis => Fast Flux (e.g., Fluxer Botnet in 2015)
  - Change the domain name on a regular basis => DGAs (e.g., Emotet malware)
  - Malware connects to a seemingly clean domain => Domain Fronting
  - Communicate over channels like Dropbox or Evernote (e.g., BKDR_VERNOT.A in 2013)
  - IRC based botnets multiplexing multiple channels (e.g., DorkBot botnet in 2015)
- Mitigations for the mitigations?.
  - E.g., partnering with registrars where the cybercriminals bought the domain names associated with the botnet (it's C2 server(s))

Fast Flux – What it is

- Basic idea: A single fully qualified domain name, where IP addresses are swapped in and out with high frequency (TTL <300s), through changing DNS records.
- Fast flux is a DNS technique to hide a machine behind an ever-changing network of compromised hosts acting as proxies.
  - Makes malware networks more resistant to IP based blocking
  - The Storm Worm (2007) was one of the first malware variants to make use of this technique.
  - It can also refer to the combination of peer-to-peer networking, distributed command and control, web-based load balancing and proxy redirection
- How to defend?
  - Block the fully qualified domain or take it down with the help of the registrars/providers where the domains were registered

Fast Flux - Visualization

DNS Robtex Analysis of a Fast flux domain



Domain Generation Algorithm (DGA)

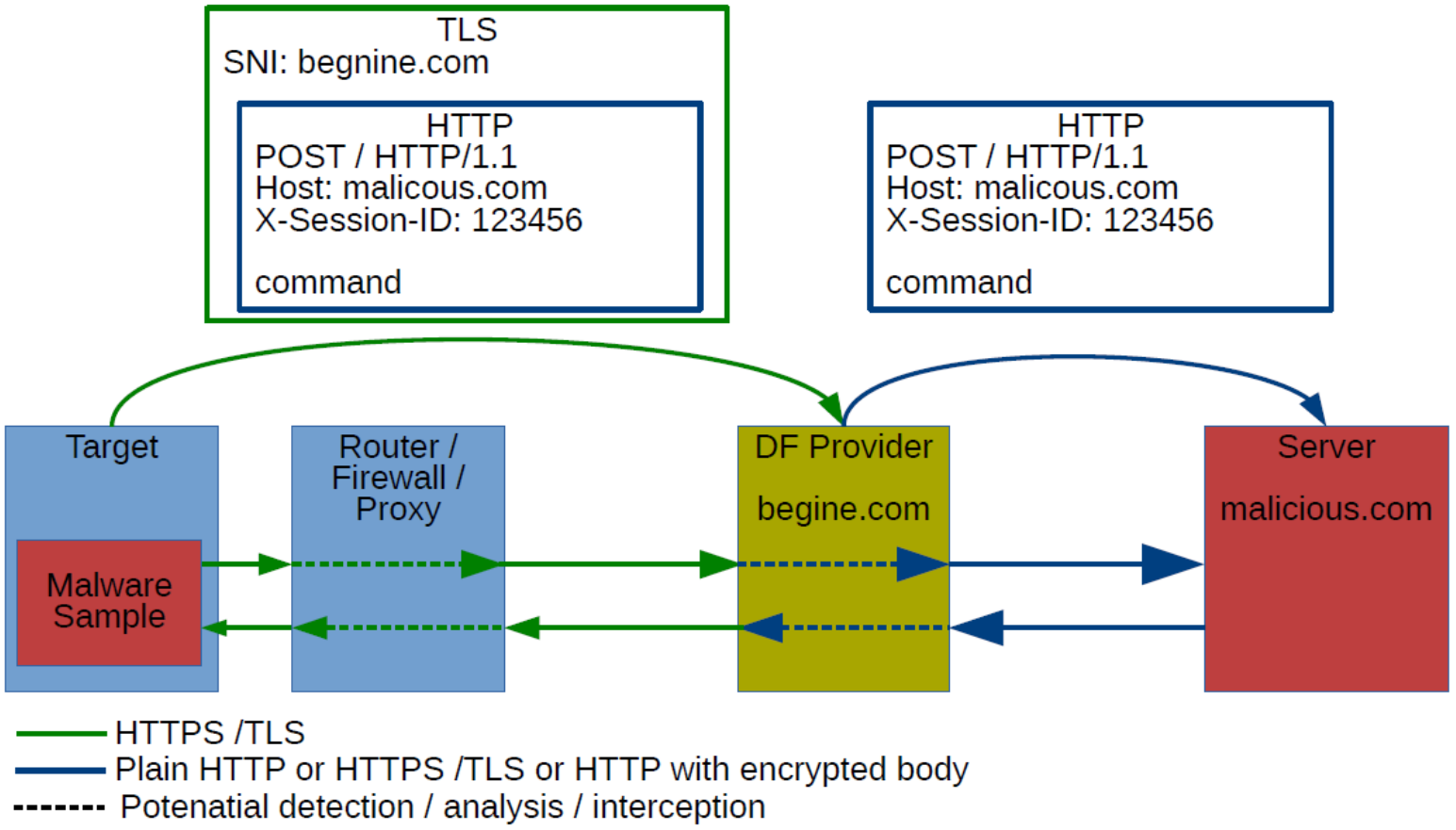
- Thwart "bad domain" list defense with domain generation algorithm
- Used to periodically generate many domain names that can be used as rendezvous points with their C2 servers.
- The large number of potential rendezvous points makes it difficult for law enforcement to effectively shut down botnets

```
def generate_domain(year, month, day):
    """Generates a domain name for the given date."""
    domain = ""

    for i in range(16):
        year = ((year ^ 8 * year) >> 11) ^ ((year & 0xFFFFFFF0) << 17)
        month = ((month ^ 4 * month) >> 25) ^ 16 * (month & 0xFFFFFFF8)
        day = ((day ^ (day << 13)) >> 19) ^ ((day & 0xFFFFFFFE) << 12)
        domain += chr(((year ^ month ^ day) % 25) + 97)

    return domain
```

Domain Fronting



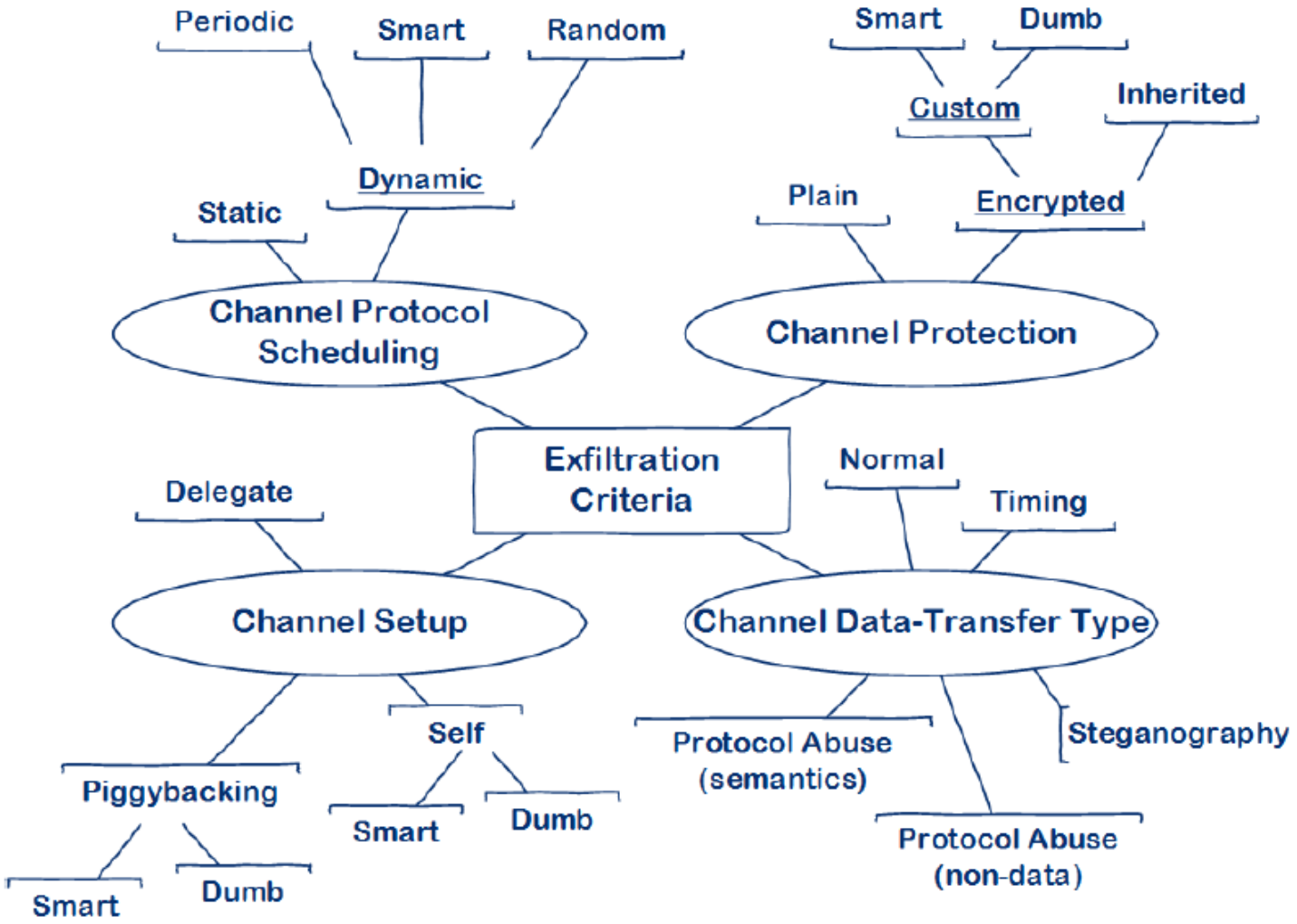
- Communicate with your C2 server using a totally legitimate domain

Malware Communication

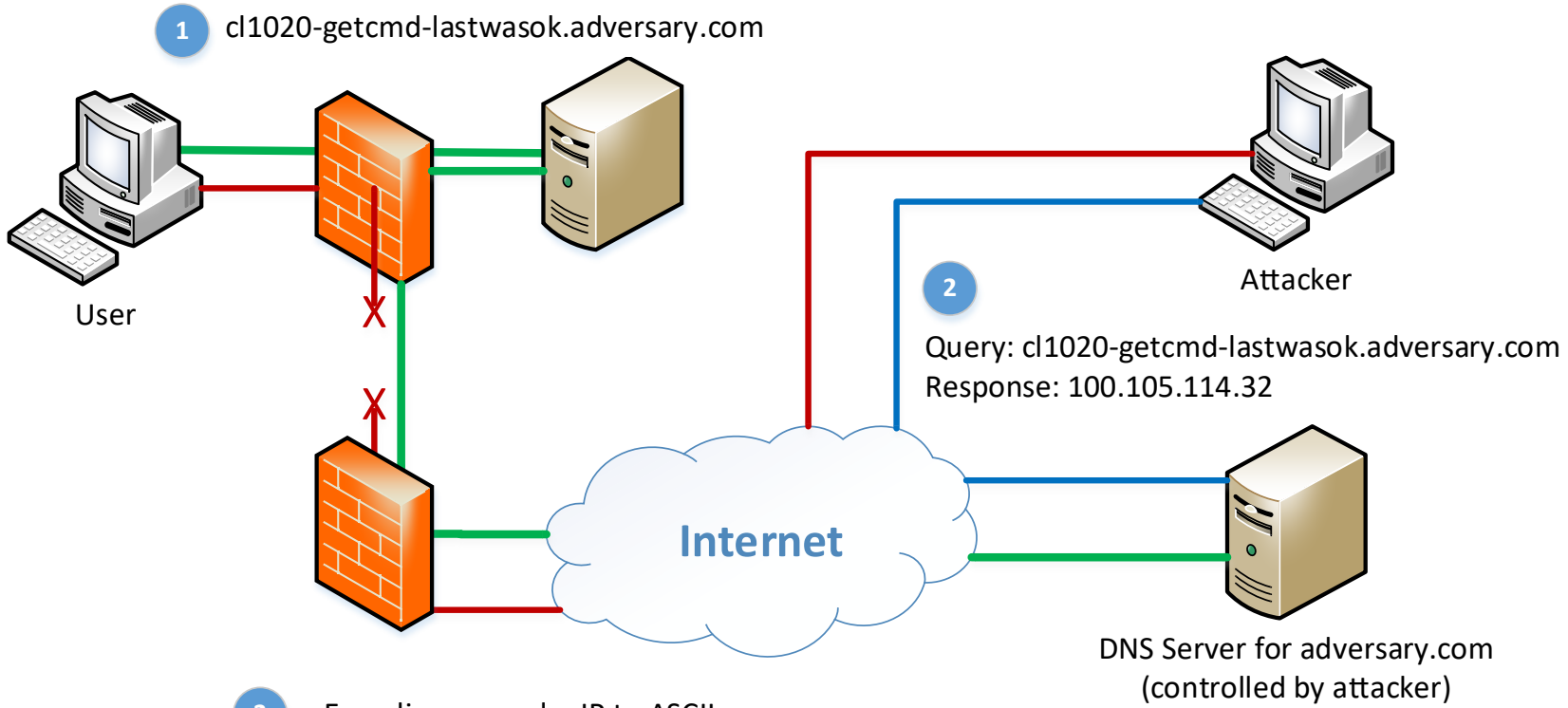
- Most common: HTTPS with or without content-level encryption
- Advanced concept: Smart communication/exfiltration
  - What protocols are used [HTTP, HTTPS, SSH, …]?
  - What applications are used [Dropbox, Google docs, Citrix, …]?
  - How much data is sent/received and when?

  => Communication should adapt to what is "normal"
- Side Channels to send out data even if communication is over an encrypted tunnel only (e.g., VPN only)
  - E.g., when machine is on WLAN, malware could delay packets to create specific packet inter-arrival patterns
- DNS Covert Channel when isolated from the Internet might work

Attributes of Communication Channels



Communication – Covert Channel using DNS

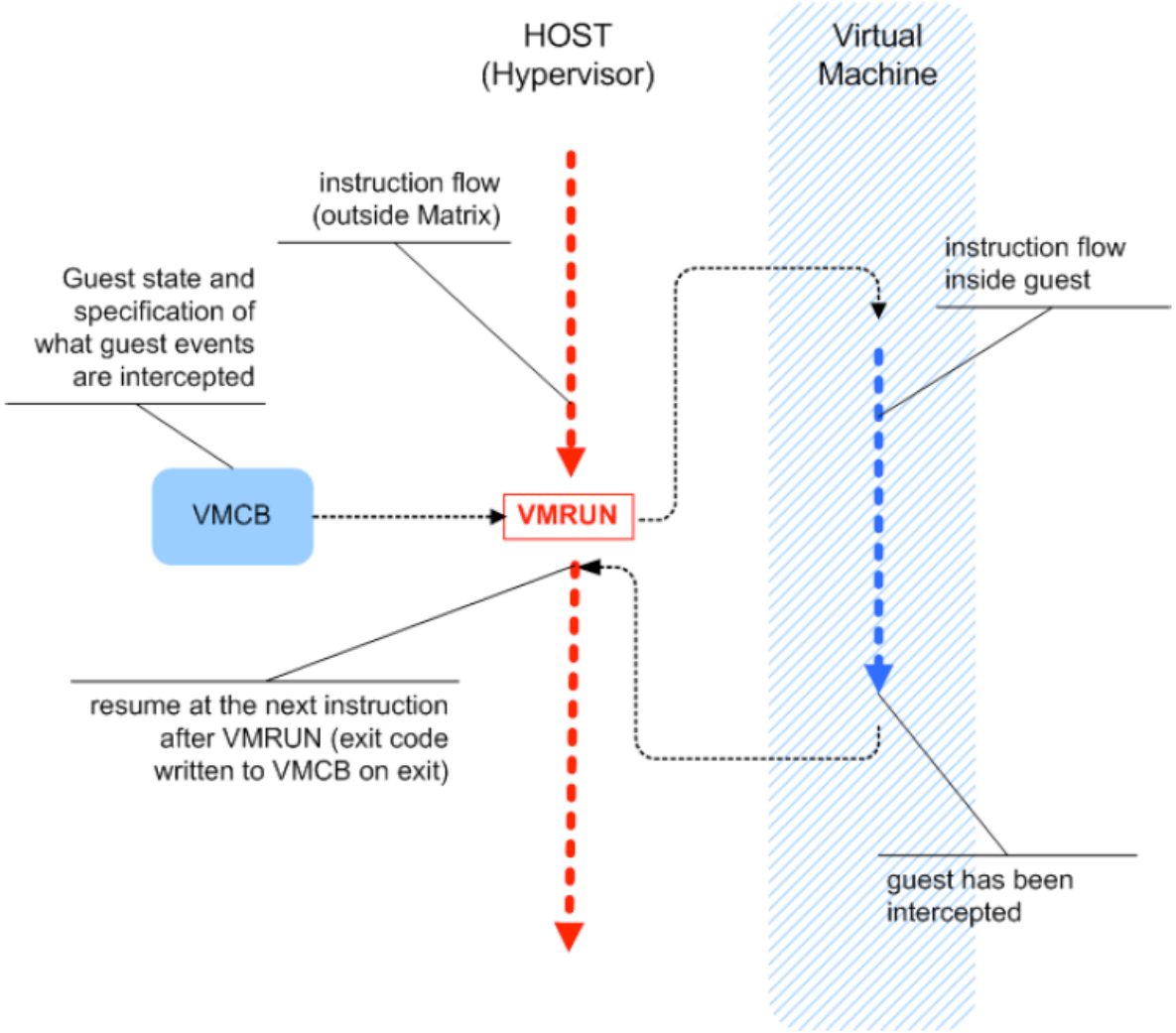


Appendix

Famous Example of a Hypervisor-level Rootkit

- Blue Pill rootkit for Windows Vista x64
- Introduced by researcher Joanna Rutkowska in 2006
- Blue Pill idea:
  - Exploit AMD64 SVM extension (later also Intel VT) to move the operating system into a virtual machine 'on-the-fly'
  - Provide thin hypervisor to control the OS
  - Hypervisor is responsible for controlling interesting events inside guest OS
- The VMRUN instruction is the heart of AMD64 SVM (and Blue Pill)

Blue Pill – The VMRUN Instruction (heart of AMD64 SVM)



Blue Pill – The Basic Idea (simplified)